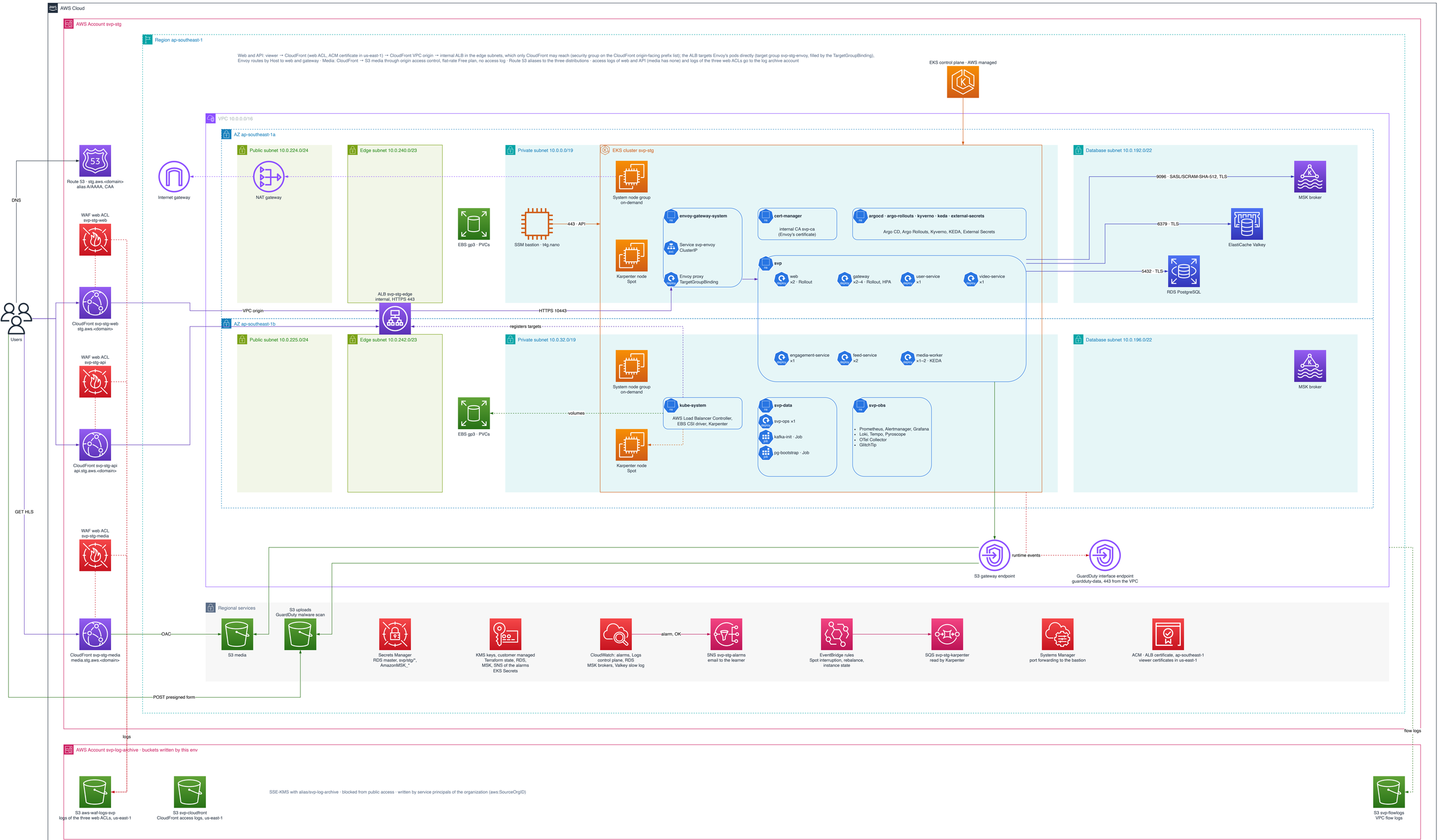
stg — EKS cluster svp-stg

ap-southeast-1 · 2 AZs, each with a public (NAT gateway only), an edge (the internal ALB only), a private and a database subnet · one NAT gateway, in the first AZ, for every private subnet · system node group on-demand and Karpenter's nodes (Spot), one icon each per AZ · API endpoint private only, reached through Systems Manager port forwarding to the SSM bastion · EBS volumes in the AZ of their pod's node · VPC flow logs, WAF logs and CloudFront access logs are written to S3 buckets of the svp-log-archive account

Database subnets, no route to the Internet: RDS single-AZ · Amazon MSK one broker per AZ, SASL/SCRAM-SHA-512 over TLS · ElastiCache for Valkey one node · AWS picks the AZs of RDS and ElastiCache (drawn from the first) · CloudWatch alarms of RDS, MSK and ElastiCache notify an SNS topic encrypted with a customer managed key, which emails the learner · KMS, Secrets Manager, CloudWatch, SNS, SQS, EventBridge regional

Kubernetes: a frame per namespace, an icon per workload with its replicas · the scheduler puts each pod on any node of any AZ, so the row of a frame or an icon says nothing about where its pods run (svp is drawn across the first two AZs to show it) · third-party tools as text · EKS add-ons (CoreDNS, kube-proxy, VPC CNI) not drawn; the GuardDuty runtime agent add-on reports through the guardduty-data interface endpoint

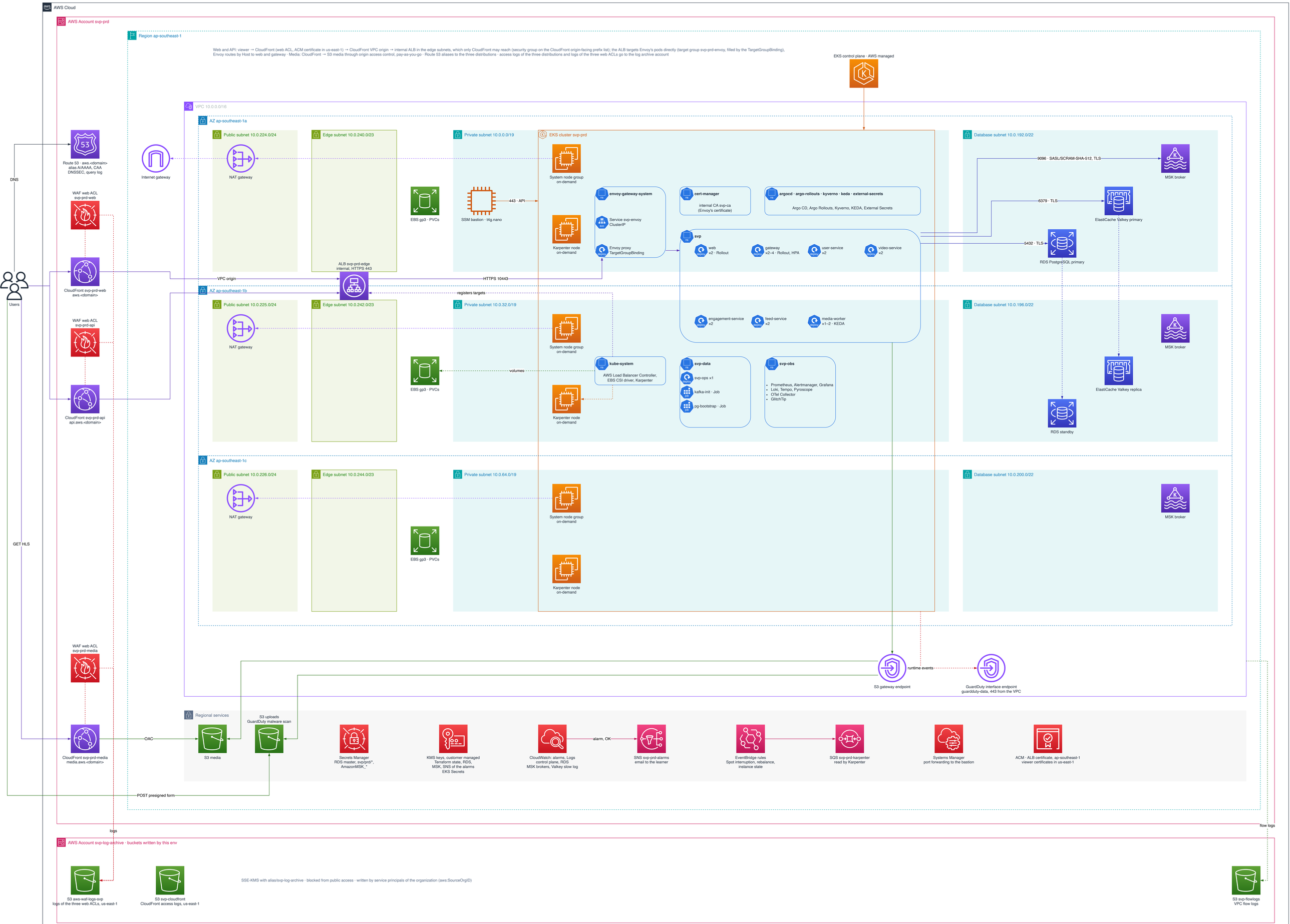


prd — EKS cluster svp-prd

ap-southeast-1 · 3 AZs, each with a public (NAT gateway only), an edge (the internal ALB only), a private and a database subnet · one NAT gateway per AZ · system node group on-demand and Karpenter's nodes (on-demand), one icon each per AZ · API endpoint private only, reached through Systems Manager port forwarding to the SSM bastion · EBS volumes in the AZ of their pod's node · VPC flow logs, WAF logs and CloudFront access logs are written to S3 buckets of the svp-log-archive account

Database subnets, no route to the Internet: RDS Multi-AZ primary/standby · Amazon MSK one broker per AZ, SASL/SCRAM-SHA-512 over TLS · ElastiCache for Valkey primary and 1 replica(s), Multi-AZ, automatic failover · AWS picks the AZs of RDS and ElastiCache (drawn from the first) · CloudWatch alarms of RDS, MSK and ElastiCache notify an SNS topic encrypted with a customer managed key, which emails the learner · KMS, Secrets Manager, CloudWatch, SNS, SQS, EventBridge regional

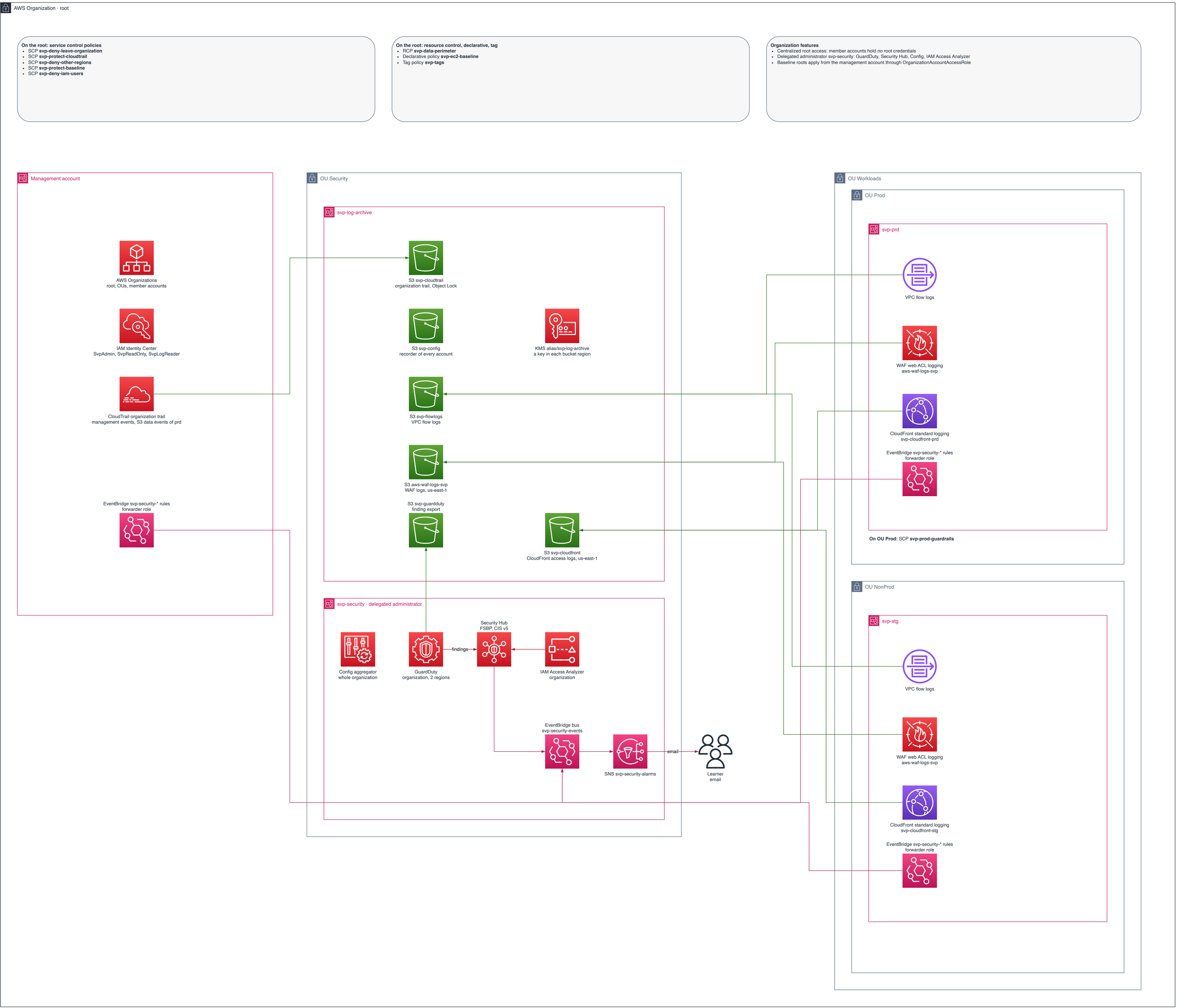
Kubernetes: a frame per namespace, an icon per workload with its replicas · the scheduler puts each pod on any node of any AZ, so the row of a frame or an icon says nothing about where its pods run (svp is drawn across the first two AZs to show it) · third-party tools as text · EKS add-ons (CoreDNS, kube-proxy, VPC CNI) not drawn; the GuardDuty runtime agent add-on reports through the guardduty-data interface endpoint



org — organization, accounts and security flows

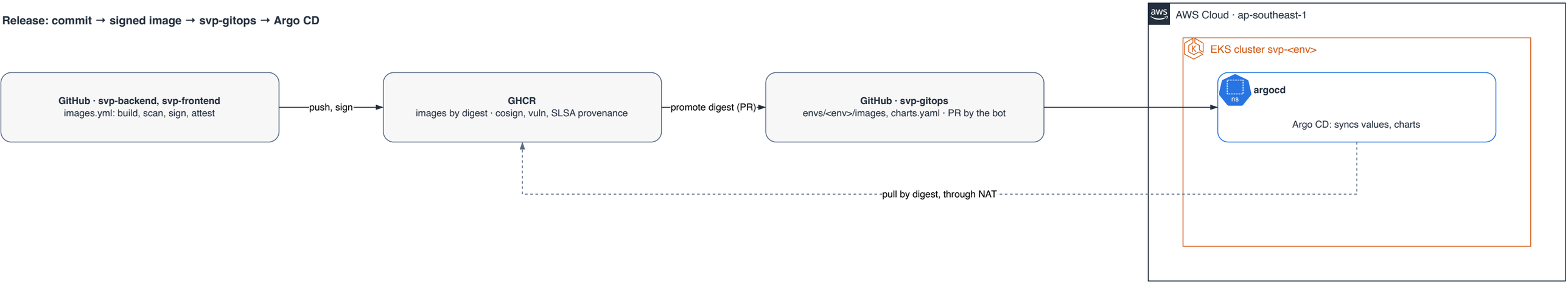
The management account holds the organization, IAM Identity Center and the organization trail · accounts in OUs as in the AWS Security Reference Architecture: Security (log archive, security tooling) and Workloads (Prod, NonProd) · every account also runs the baseline (S3 Block Public Access, EBS default encryption, Config recorder to svp-config)

Green: logs to the log archive account · red: detection and events to the security account, then SNS and email · policies on the root reach every member account (SCP, RCP) or every account (declarative, tag)

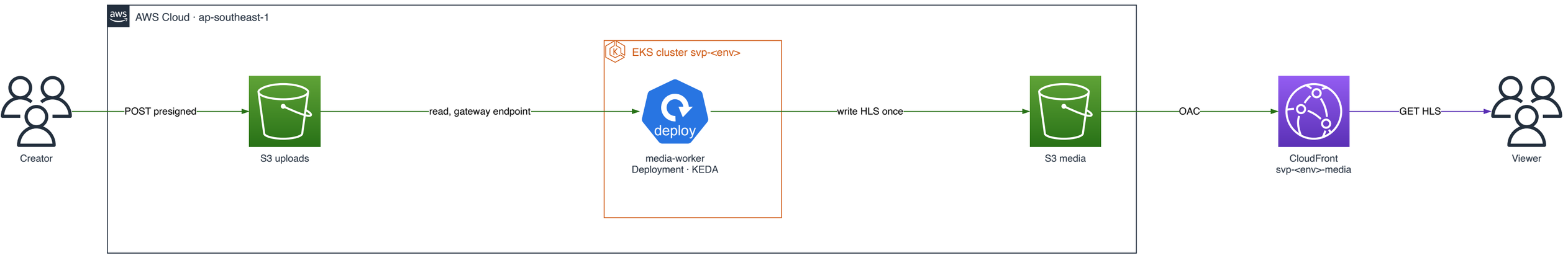


Flows — release path, upload to playback and request to the web and API

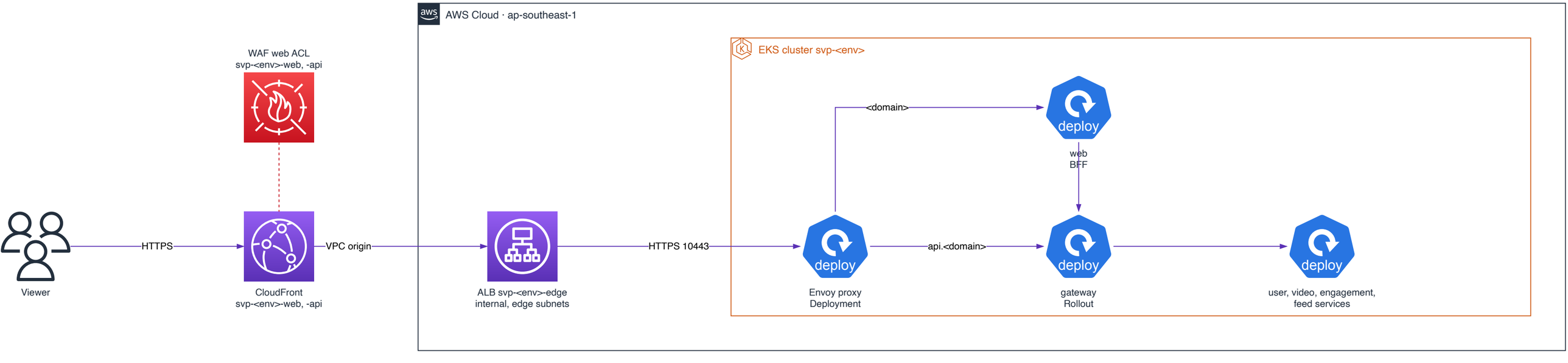
Release: commit → signed image → svp-gitops → Argo CD



Upload to playback



Request to the web and API



CloudFront adds CloudFront-Viewer-Address and the gateway trusts it only on connections from the VPC; web forwards it to the gateway · the ALB accepts only CloudFront (security group on the origin-facing prefix list) · access logs of the distributions (none for media on the flat-rate Free plan) and the web ACL logs go to the log archive account